Hydraulic Press: Operation and Maintenance Manual

Applies to: Schuler HP-400, Lasco SPS 250, Dieffenbacher DCP-630. Document no. OM-HYDRAULIC-PRESS-2026.

1. Safety

Lock out and tag out all energy sources before any maintenance. Only trained personnel may work on this machine. Wear the PPE required by the plant safety rules.

2. Sensors and alarm thresholds

An alarm is raised when a value stays above the threshold for at least two consecutive samples. HIGH is a warning; HIGH_HIGH is critical.

Sensor	Location	Unit	Normal	HIGH (warning)	HIGH_HIGH (critical)
pressure	hydraulic circuit	bar	approx. 180	> 220	> 250
temperature	oil tank	degC	approx. 45	> 65	> 75
flow	pump outlet	l/min	approx. 90	> 120	> 140
vibration	ram	mm/s	approx. 2.2	> 4.8	> 7.5

3. Alarm response

HIGH (warning): Acknowledge within 15 minutes. Review the trend of the last 2 hours and schedule an inspection within 24 hours. Follow the troubleshooting table for the sensor.

HIGH_HIGH (critical): Reduce load or stop the machine within 5 minutes and notify the shift lead. Open a corrective maintenance order. Do not restart until the root cause is found.

COMM_FAULT: Sensor readings are invalid (quality 'bad'). Check the sensor cable, connector and I/O module. If the fault lasts longer than 4 hours, take manual readings every hour.

4. Troubleshooting

4.1 High pressure (hydraulic circuit, bar)

Probable causes:

- Pressure relief valve set incorrectly
- Blocked return line or return filter
- Proportional valve fault

Corrective actions:

1. Check the main relief valve setting: 260 bar. Only authorised staff may adjust it.
2. Inspect the return filter indicator; replace element SP-HYD-RF-10 when red.
3. Run the proportional valve self-test from the HMI service menu.

4.2 High temperature (oil tank, degC)

Probable causes:

- Oil cooler fouled or cooling water flow too low
- Internal leakage in cylinders or valves
- Low oil level in the tank

Corrective actions:

1. Normal oil temperature is 40-55 degC; check cooling water flow (min 30 l/min).
2. Check for internal leakage by monitoring pressure decay with the ram held at top position.
3. Above 75 degC the oil degrades: after any HIGH_HIGH event take an oil sample and replace the oil (HLP 46, 400 l) if the analysis fails.

4.3 High flow (pump outlet, l/min)

Probable causes:

- Pump displacement controller fault
- Leak in the pump outlet line
- Accumulator pre-charge lost

Corrective actions:

1. Check the pump displacement controller signal (4-20 mA).
2. Inspect hoses and fittings on the pump outlet; replace hoses every 6 years.
3. Check accumulator nitrogen pre-charge: 90 bar.

4.4 High vibration (ram, mm/s)

Probable causes:

- Worn ram guides or gibs
- Die misalignment
- Loose tooling clamps

Corrective actions:

1. Measure guide clearance; adjust gibs if clearance exceeds 0.1 mm.
2. Re-align the die using the alignment pins.
3. Check hydraulic tool clamps for full clamping pressure.

5. Maintenance schedule

Interval	Task
Daily	Check oil level and leaks
Every 30 days	Check accumulator pre-charge (90 bar)
Every 6 months	Oil analysis (ISO 4406 target 17/15/12)
Every 6 years	Replace all hydraulic hoses